

pregnancy-reminder-bot

Technical documentation

1. Overview

A WhatsApp bot that reminds pregnant women to take their daily FE (iron) tablet, tracks day by day adherence, validates delivery around the due date, and reminds about the four standard postpartum visits (KF/KN). A local admin web dashboard manages users, quotas, alarms, and data export. The bot runs on the WhatsApp Web protocol through whatsapp-web.js and stores everything in SQLite.

2. Components

Component	Role
whatsapp-web.js session	Connects a WhatsApp number. QR code or 8-digit pairing code.
Scheduler loop	Runs every 30 s. Decides which reminder is due for which user.
Send guard	Single outbound queue. Enforces random gaps, per-minute/hour/day quotas, warm-up curve, and a circuit breaker. See section 6.
Conversation handlers	Parse user answers: onboarding, polls, delivery data, commands.
Admin web (Express)	Loopback-only HTTP dashboard on port 3030 with login, settings, health, alarms, per-user actions, CSV export.
SQLite	users, reminder_logs, postpartum_visit_logs. WAL mode. Daily backup via VACUUM INTO.

System overview

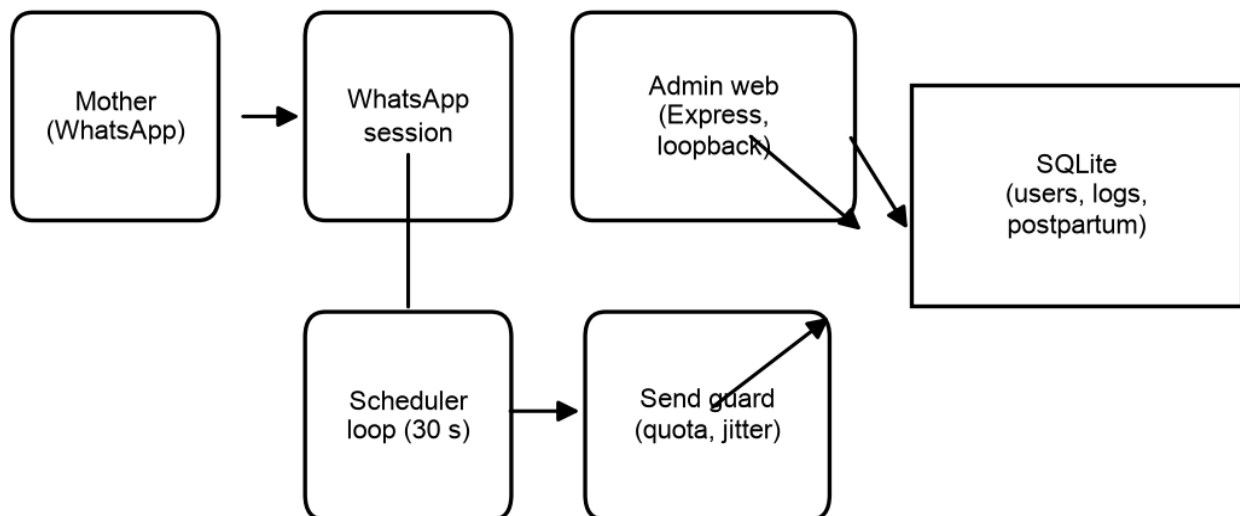


Figure 1. System overview. Every outbound message passes the send guard.

3. User flows

3.1 Onboarding

A new user answers nine questions: name, age, pregnancy number, HPHT date, routine medicine use (yes/no), tea consumption (yes/no), who usually reminds them, consent to be reminded (yes/no), and reminder time in 24h format. Each answer is validated; a malformed value (wrong date format, unclear time, non yes/no reply) re-asks the same question. When onboarding finishes the user becomes active and the daily FE reminder starts.

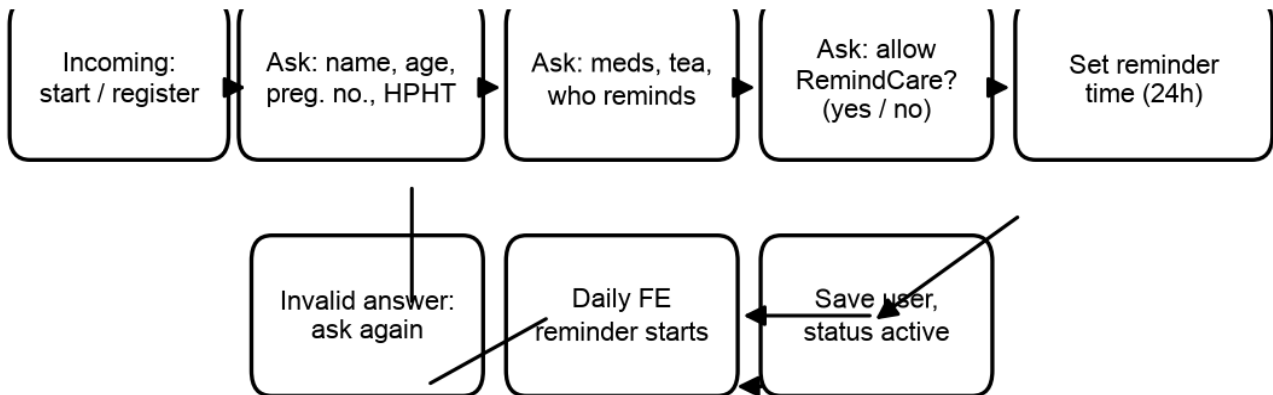


Figure 2. Onboarding flow.

3.2 Daily FE reminder

Every 30 s the scheduler checks active users. A reminder fires only inside the send window (06:00-21:30 by default), exactly at the user's reminder time, and only once per day. The bot sends a poll with Sudah / Belum options and waits up to 30 minutes for the answer. An unanswered poll counts as one attempt, up to 4; after that the day is marked missed and is never re-sent. This rule keeps a restart from flooding every user with overdue reminders.

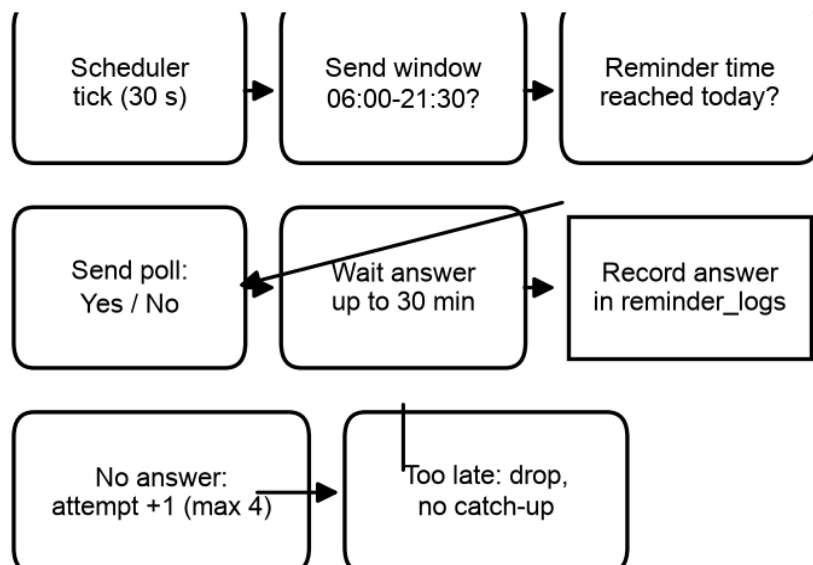


Figure 3. Daily FE reminder flow.

3.3 Delivery validation

From week 39 (configurable via DELIVERY_VALIDATION_START_WEEK) the bot asks daily whether the mother has given birth. A 'No' answer keeps the poll running. A 'Yes' answer opens the delivery data form: date, time, place, birth attendant, complications, baby gender, and birth weight. When the data is saved the user is marked completed and postpartum scheduling begins.

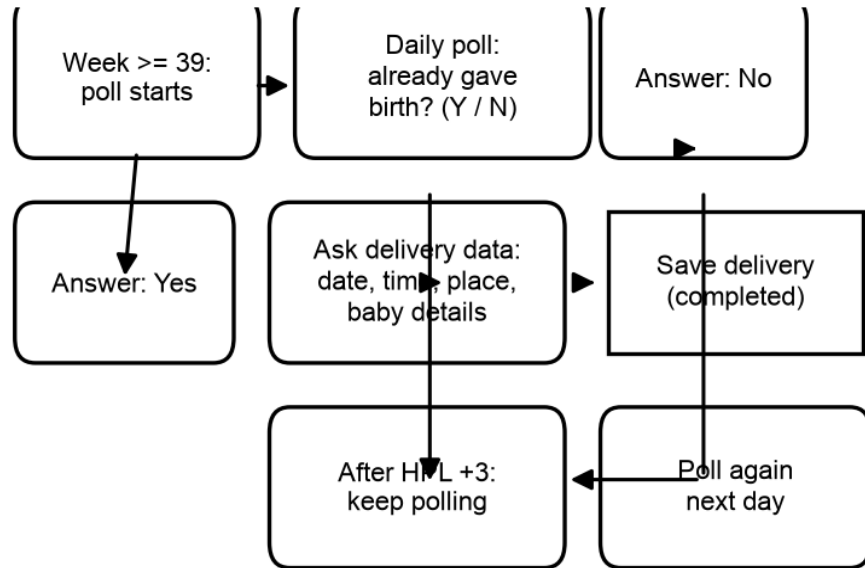


Figure 4. Delivery validation flow.

3.4 Postpartum visits

After delivery the bot schedules four visits: KFKN1 (6-48 h), KFKN2 (3-7 d), KFKN3 (8-28 d), and KF4 (29-42 d) after birth. When a window opens, the bot sends one visit poll and logs the answer. At most one visit poll is sent per scheduler cycle per user, so the chat is never flooded. The program finishes after all four visits are answered.

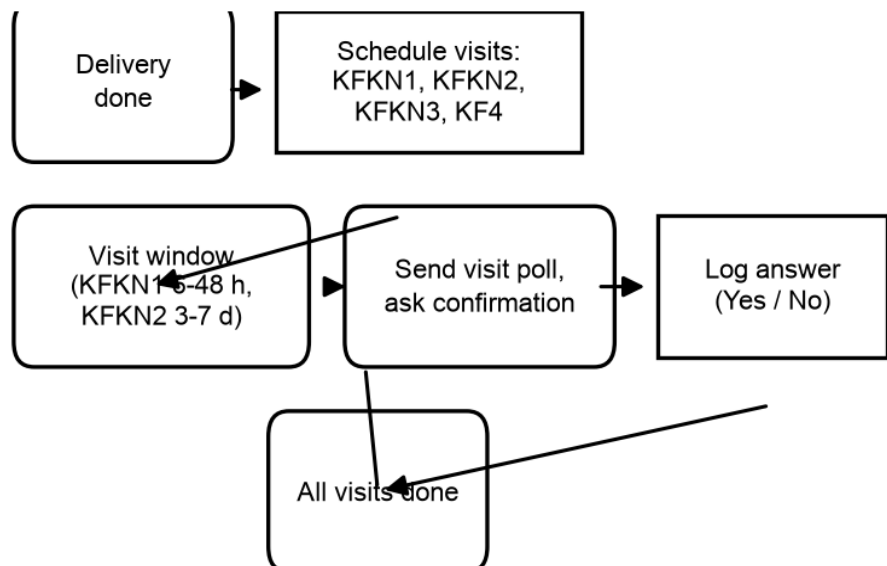


Figure 5. Postpartum visit flow.

3.5 Admin web

The admin web binds to 127.0.0.1:3030 only. Login is rate limited (5 failed attempts block the IP for 15 minutes). After login the admin sees a dashboard with user summary, health status, alarms, and quota usage, then can open a user detail page with logs, delivery data, and postpartum visits, and act on a user (pause, reset, emergency stop) or export the data as CSV.

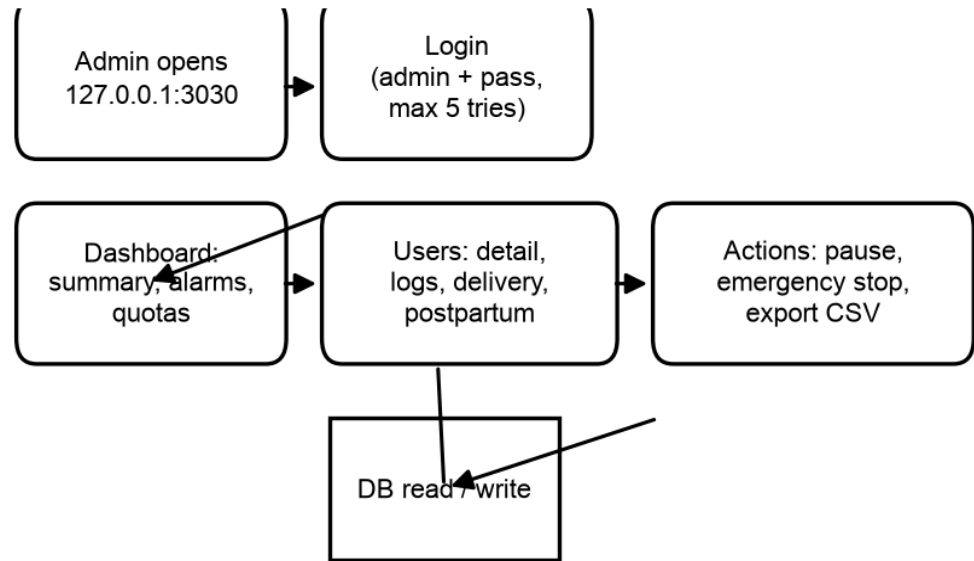


Figure 6. Admin web flow.

4. Confidence and safety

Outbound volume is intentionally low to stay under WhatsApp automation limits. The send guard adds a random gap of 3.5-6 s between messages and a reply gap of 0.9-2.1 s, caps outbound traffic at 12/min, 180/h, 900/day, and applies a 30-day warm-up curve that starts new sessions at 50 messages a day. Five consecutive permanent failures open a circuit breaker that pauses sending for one hour. Stale reminders (older than 30 minutes) are dropped instead of re-sent.

Parameter (default)	Value	Purpose
SEND_MIN_GAP_MS / SEND_JITTER_MS	3500 / 2500	Random 3.5-6 s gap between sends
SEND_MAX_PER_MINUTE / HOUR / DAY	12 / 180 / 900	Hard outbound quotas
SEND_WARMUP_DAY1_MAX / DAY8_MAX	50 / 150	Warm-up curve for new sessions
SEND_FAIL_BREAKER / SEND_PAUSE_MS	5 / 3600000	Circuit breaker, 1 h pause
REMINDER_STALE_AFTER_MINUTES	30	Drop overdue reminders
SEND_WINDOW_START_HOUR / END_HOUR	6 / 21:30	Scheduled reminder window

5. Deployment notes

The repo ships systemd units (bot service and daily backup timer at 03:30) in `deploy/`. The backup script runs `VACUUM INTO`, checks `PRAGMA integrity_check`, and keeps backups for 14 days. The bot prefers a real installed Chrome and rejects invalid numeric env values on start, so a misconfigured safety control never silently disables itself. Session folders (`.webjs_auth`) and the data directory are gitignored; never commit them.

6. Related documents

`deploy/INSTALL.md`: full server install, pairing, and restore procedures. `README.md`: quick start. `.env.example`: complete configuration reference.

Version 1.0, September 2026. This document describes the public repository at github.com/azrahudaya/pregnancy-reminder-bot.